DOSAGE CALCULATION #3

NUR3260

WEIGHT BASED MEDICATIONS

- Many medications are prescribed by weight – especially in pediatric population
  - Basing medications on weight ensures correct dosages

MG/KG/DAY

- Some prescriptions may be written as milligrams per kilogram of body weight per day
- We then need to determine how many milligrams of a medication the client can have per day based on their weight in kilograms

Order states: Administer 20mg/kg/day
Client weighs 112lbs
How many milligrams can they have per day?

1) need to convert 112lbs to kg

112lb/2.2 = 50.9kg

2) take mg x kg

20mg x 50.9kg = 1,018mg/day

OR

112lb	1 kg	20mg	2240	= 1,018mg/day
	2.2lb	1kg	2.2lb	

SINGLE DOSES

- prescription should then indicate how many doses the client will receive throughout the day
- Most often every 4, 6, 8 , or 12 hours (q4, q6, etc…)
- Divide total daily by how many times a day client will receive…so:
  - Q4 = 6 times a day (24 hours divided by every 4 hours = 6)
  - Q6 = 4 times a day
  - Q8 = 3 times a day
  - Q12 = 2 times a day

- So example from last slide = 1,018mg/day
- How many mg would be delivered per dose if ordered Q6? What about Q12? Round to nearest whole number

Q6 = 1,018/4 = 255mg

Q12 = 1,018/2 = 509mg

If doing dimensional analysis you can add this into the equation:
- add 1 day over 24 hours and ___ hours over 1dose

PRACTICE

Provider orders Tobramycin 4mg/kg/day to be administered every 8 hours

Client weighs 60kg

Available: Tobramycin 40mg/2ml

How many ml should the nurse give per dose? (round to whole number)

$$\frac{60kg}{} \quad \frac{4mg}{1kg} \quad \frac{2ml}{40mg} \quad \frac{1\ day}{24\ hours} \quad \frac{8\ hour}{1\ dose} = \frac{3840}{960} = 4ml$$

OR

4mgx 60 = 240mg per day

$$\frac{240mg}{40mg} \quad X\ 2ml = 12ml\ per\ day \qquad 12 / 3 = 4ml\ per\ dose$$

MORE PRACTICE

Order states: administer lorazepam 0.05mg/kg one time before surgery.

Client weighs 192lbs

Available: Lorazepam 2mg/ml

How many ml will the nurse administer? Round to nearest whole number

192lb	1kg	0.05mg	1ml	9.6	= 2.09 = 2 ml
	2.2lb	1kg	2mg	4.4	

OR

192/2.2 = 87.272727 kg

0.05 x 87.272727 kg = 4.363636mg

4.363636	x1ml	= 2ml
2		

WEIGHT BASED PRACTICE

You have on hand hydromorphone (Dilaudid) 4 mg/mL. You need to administer 0.015 mg/kg to a patient who weighs 150 pounds. How many milliliters should you administer? **Record your answer using two decimal places.**

1)

150lbs	=	
2.2	68.2kg	

2) 68.2kg x 0.015mg/kg = 1.023 mg

3)

1.023mg	1mL	= 0.26mL
	4mg	

OR

150lb	1kg	0.015mg	1mL	2.25	= 0.255 = 0.26mL
	2.2lb	1kg	4mg	8.8	

CONSIDERATIONS

- What needs to be in prescription
  - Date and time prescription written
  - Client's full name
  - Name of medication
  - Indication (reason)
  - Dose: amount times or frequency of administration and strength
  - Route of admin
  - Any specific instructions (ex. Take with food or without)
  - Signature
- Abbreviations to avoid:
  - QD – once daily but looks like QID (4 times daily)
  - HS – could mean bedtime or half-strength
  - DC – could mean discharge or discontinue
- What would you do if you have a prescription missing a piece or with one of the above abbreviations?

READING A MEDICATION PACKAGE

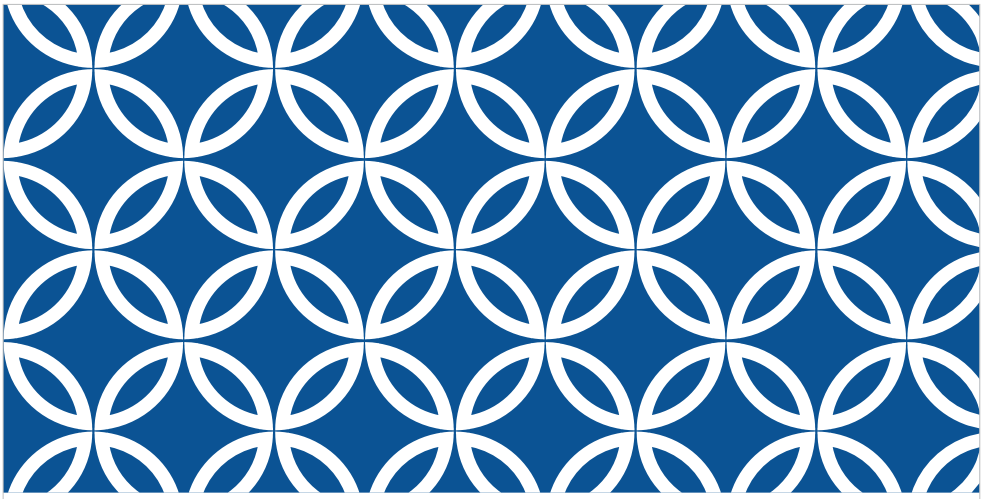


READING PACKAGING





QUESTIONS?



DOSAGE CALCULATION #3

NUR3260

WEIGHT BASED MEDICATIONS

- Many medications are prescribed by weight – especially in pediatric population
  - Basing medications on weight ensures correct dosages

MG/KG/DAY

- Some prescriptions may be written as milligrams per kilogram of body weight per day
- We then need to determine how many milligrams of a medication the client can have per day based on their weight in kilograms

Order states: Administer 20mg/kg/day
Client weighs 112lbs
How many milligrams can they have per day?

1) need to convert 112lbs to kg

112lb/2.2 = 50.9kg

2) Take mg x kg

20mg x 50.9kg = 1,018mg/day

OR

112lb	1 kg	20mg	2240	=
	2.2lb	1kg	2.2lb	1,018mg/day

SINGLE DOSES

- prescription should then indicate how many doses the client will receive throughout the day
- Most often every 4, 6, 8 , or 12 hours (q4, q6, etc…)
- Divide total daily by how many times a day client will receive…so:
  - Q4 = 6 times a day (24 hours divided by every 4 hours = 6)
  - Q6 = 4 times a day
  - Q8 = 3 times a day
  - Q12 = 2 times a day

- So example from last slide = 1,018mg/day
- How many mg would be delivered per dose if ordered Q6? What about Q12? Round to nearest whole number

Q6 = 1,018/4 = 255mg

Q12 = 1,018/2 = 509mg

If doing dimensional analysis you can add this into the equation:
- add 1 day over 24 hours and ___ hours over 1dose

PRACTICE

Provider orders Tobramycin 4mg/kg/day to be administered every 8 hours

Client weighs 60kg

Available: Tobramycin 40mg/2ml

How many ml should the nurse give per dose? (round to whole number)

$$\frac{60\text{kg}}{1\text{kg}} \times \frac{4\text{mg}}{1\text{day}} \times \frac{2\text{ml}}{40\text{mg}} \times \frac{24\text{ hours}}{8\text{ hour}} \times \frac{1\text{ dose}}{1} = \frac{3840}{960} = 4\text{ml}$$

OR

4mgx 60 = 240mg per day

$$\frac{240\text{mg}}{40\text{mg}} \times 2\text{ml} = 12\text{ml per day}$$

12 / 3 = 4ml per dose

MORE PRACTICE

Order states: administer lorazepam 0.05mg/kg one time before surgery.

Client weighs 192lbs

Available: Lorazepam 2mg/ml

How many ml will the nurse administer? Round to nearest whole number

192lb	1kg	0.05mg	1ml	9.6	= 2.09 = 2 ml
	2.2lb	1kg	2mg	4.4	

OR

192/2.2 = 87.272727 kg

0.05 x 87.272727 kg = 4.363636mg

4.363636	x1ml	= 2ml
2		

WEIGHT BASED PRACTICE

You have on hand hydromorphone (Dilaudid) 4 mg/mL. You need to administer 0.015 mg/kg to a patient who weighs 150 pounds. How many milliliters should you administer? **Record your answer using two decimal places.**

1) $\frac{150\text{lbs}}{2.2} = 68.2\text{kg}$

2) $68.2\text{kg} \times 0.015\text{mg/kg} = 1.023\text{ mg}$

3) $\frac{1.023\text{mg}}{} \quad \frac{1\text{mL}}{4\text{mg}} = 0.26\text{mL}$

OR

$\frac{150\text{lb}}{} \quad \frac{1\text{kg}}{2.2\text{lb}} \quad \frac{0.015\text{mg}}{1\text{kg}} \quad \frac{1\text{mL}}{4\text{mg}} \quad \frac{2.25}{8.8} = 0.255 = 0.26\text{mL}$

CONSIDERATIONS

- What needs to be in prescription
  - Date and time prescription written
  - Client's full name
  - Name of medication
  - Indication (reason)
  - Dose: amount times or frequency of administration and strength
  - Route of admin
  - Any specific instructions (ex. Take with food or without)
  - Signature
- Abbreviations to avoid:
  - QD – once daily but looks like QID (4 times daily)
  - HS – could mean bedtime or half-strength
  - DC – could mean discharge or discontinue
- What would you do if you have a prescription missing a piece or with one of the above abbreviations?

READING A MEDICATION PACKAGE



READING PACKAGING




QUESTIONS?